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SUBJECT CODE NO: - RNP-8006
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III
Examination November / December 2025 (To be held in February-2026)
Data Structures

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from question no's 2, 3, 4, 5, 6 and 7

SECTION-A

Q1 **Solve any TEN from the following questions** **20**

a) Define pointer and give its uses in data structures.

b) What is dynamic memory allocation?

c) Define Big Θ notation.

d) Write postfix expression for: (A + B) * (C-D).

e) Define deque and give one application.

f) What is a doubly linked list?

g) Define binary search tree.

h) What is heap sort?

i) Define collision in hashing.

j) What is adjacency matrix?

k) Write the difference between stack and queue.

l) Define header linked list.

m) What is meant by depth of a binary tree?

SECTION-B

Q2 a) Explain abstract data type with a suitable example. **05**

b) Explain conversion of infix to postfix expression using a stack. **05**

Q3 a) Describe the concept of circular queue with algorithm. **05**

b) Explain insertion and deletion operations in doubly linked list. **05**

Q4 a) Explain binary tree representation in memory. **05**

b) Write an algorithm for deleting a node from a binary search tree. **05**

Q5 a) Explain quick sort with example. **05**

b) Compare and analyse merge sort and quick sort based on complexity. **05**

Q6 a) Write an algorithm for sequential and binary search. **05**

b) Explain hash table with an example. **05**

Q7 a) Explain Dijkstra's shortest path algorithm with example. **05**

b) Write a short note on spanning trees. **05**